

ADIA : Lab

# **Investment Lessons from Cosmology: “Draw Your Assumptions Before Your Conclusions”**

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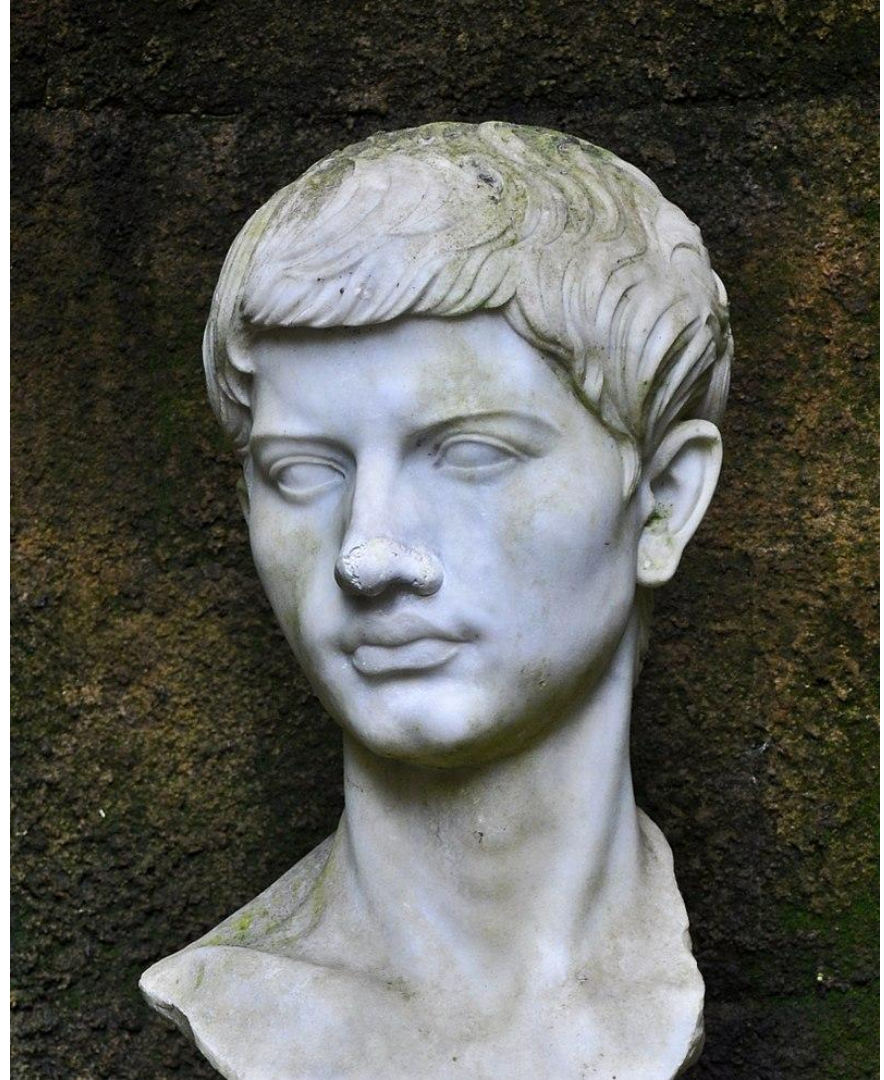
# Seminar's Objective

- Every student of statistics learns that correlation does not imply causation
  - Association is an observational property
  - Causation is an interventional concept
- Causality plays a fundamental role in the scientific method
  - Scientific theories are falsifiable statements of the form “ $X$  causes  $Y$  through mechanism  $M$ ”
- Much of modern scientific research is false due to the “replication crisis”:
  - **Type-A Spuriousity: Statistical flukes**
  - **Type-B Spuriousity: Non-causal association**
- **Recent developments in Cosmology demonstrate the power of Causal Inference to prevent false discoveries**
  - Similarities between Cosmology and Financial Research provide lessons about what (and what not) to do when developing investment strategies
  - In the words of [Prof. Miguel Hernán](#), “draw your assumptions before your conclusions”

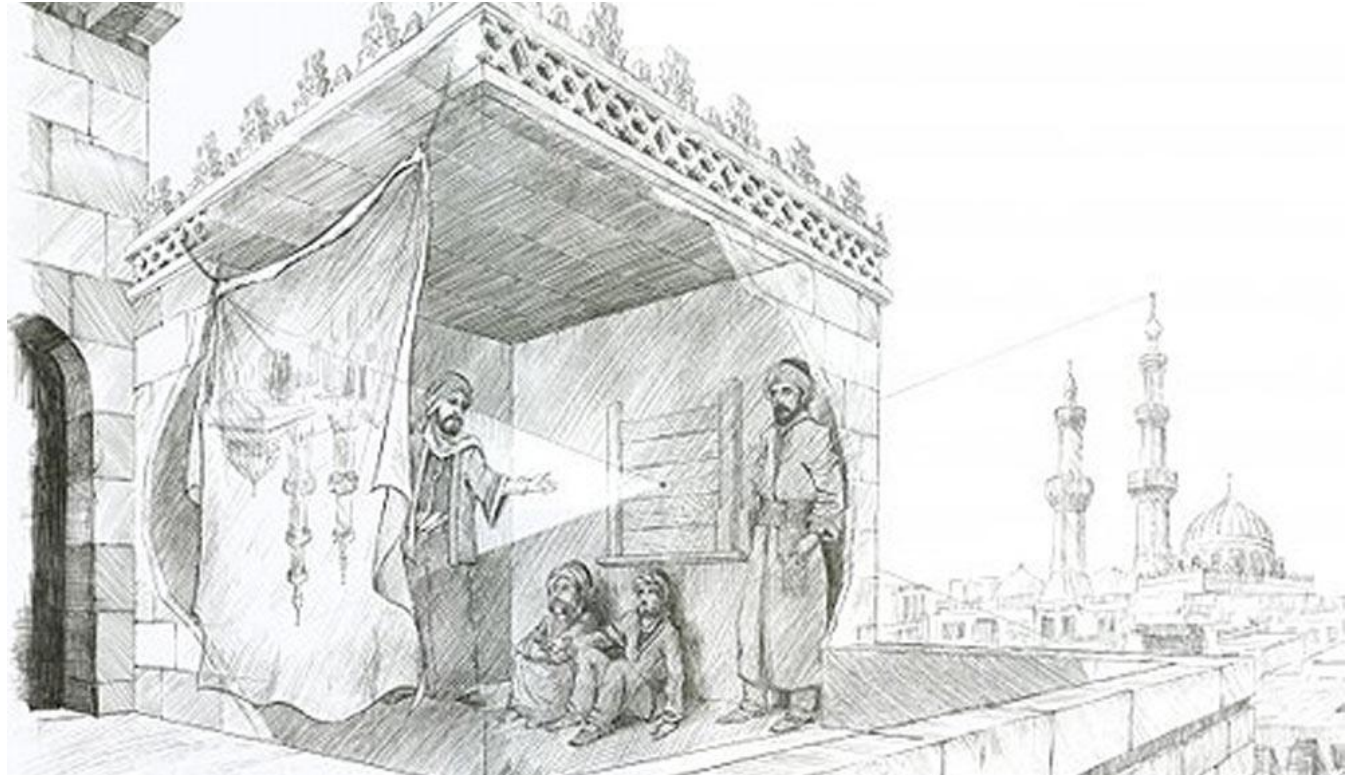
# **Why Study Cause and Effect?**

*“Happy the man, who,  
studying Nature’s laws,  
thro’ known effects can  
trace the secret cause”*

The Second Book of the Georgics  
Publius Vergilius Maro, “Virgil” (70 – 19 BC)



Around the year 1011 that Arab mathematician Hasan Ibn Al Haytham (965 - 1040) proposed a scientific method for deducing causal mechanisms.



David Hume defined a cause-effect (causal) relation as that “where, if the first object had not been, the second never had existed.”

An Enquiry concerning Human  
Understanding. Sec. VII. (1748)  
David Hume (1711 - 1776)



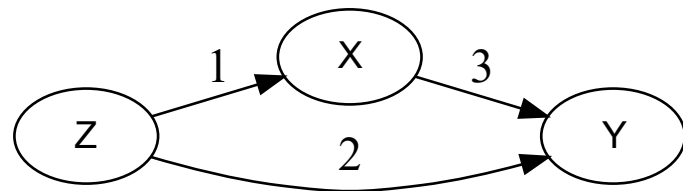
# The Three Stages of the Scientific Method

| Stage                                             | Statement                                                                                                                                                                                                    | Example                                                                |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Phenomenological</b><br>(induction)            | $X$ is associated with $Y$                                                                                                                                                                                   | CO <sub>2</sub> levels are associated with global temperature          |
| <b>Theoretical</b><br>(abduction)                 | $X$ causes $Y$ through mechanism $M$                                                                                                                                                                         | CO <sub>2</sub> causes warming via radiative forcing                   |
| <b>Falsification</b><br>Refutation<br>(deduction) | Refutation attempts: <ul style="list-style-type: none"><li>• <math>X</math> does not cause <math>Y</math></li><li>• <math>X</math> causes <math>Y</math>, but not through mechanism <math>M</math></li></ul> | Test through volcanic-eruption or solar-irradiance natural experiments |

# **Association vs. Causation**

# Paths

- A **data-generating process** can be represented as a directed acyclic graph (DAG)
  - Nodes are **variables**
  - Arrows indicate the **direction of dependence**
- A **path** is a sequence of arrows and nodes that connect two variables  $X$  and  $Y$ , regardless of the direction of causation
- A **directed path** is a path where all arrows point in the same (forward) direction
- In a directed path that starts in  $X$  and ends in  $Y$ 
  - $X$  is an **ancestor** of  $Y$ , and
  - $Y$  is a **descendant** of  $X$



A DAG is a directed graph with no directed cycles.

In the DAG above, there are two paths between  $X$  and  $Y$ :

- a) A directed path:  $X \rightarrow Y$
- b) A non-directed path:  $X \leftarrow Z \rightarrow Y$

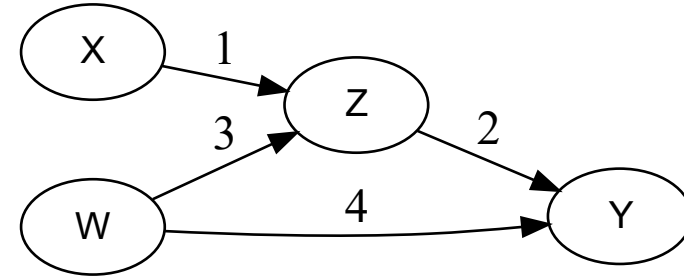
In the DAG above,  $X$  is a descendant of  $Z$ , and  $Y$  a descendant of  $X$  and  $Z$ .

# Blocked Paths

- In a DAG with three variables  $\{X, Y, Z\}$ , a variable  $Z$  is
  - a **confounder** when the causal relationships include a structure  $X \leftarrow Z \rightarrow Y$
  - a **collider** when the causal relationships are reversed, i.e.  $X \rightarrow Z \leftarrow Y$
  - a **mediator** when the causal relationships include a structure  $X \rightarrow Z \rightarrow Y$

A path between  $X$  and  $Y$  is **blocked** if either:

- a) the path traverses a collider, and the researcher has not conditioned on that collider or its descendants; or
- b) the researcher conditions on a variable in the path between  $X$  and  $Y$ , where the conditioned variable is not a collider



In the above DAG:

- $W$  is a confounder to  $Z$  and  $Y$
- $Z$  is a collider to  $X$  and  $W$
- $Z$  is a mediator between  $X$  and  $Y$

The path  $X \rightarrow Z \leftarrow W \rightarrow Y$  is blocked by  $Z$ .

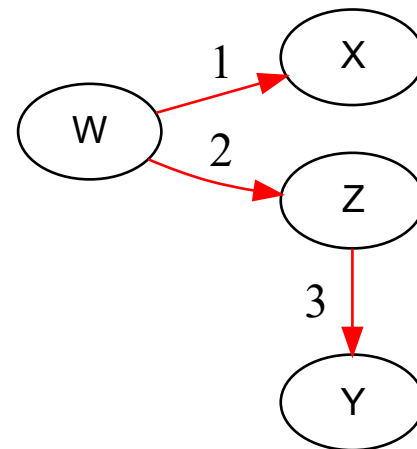
The only unblocked path between  $X$  and  $Y$  is the causal path,  $X \rightarrow Z \rightarrow Y$ .

# What is Association?

- **Association** flows along an unblocked path between  $X$  and  $Y$ 
  - Association is *symmetric* because paths do not follow the direction of causation
- Probabilistically, two variables  $X$  and  $Y$  are associated when knowing the value of one conveys information about the value of the other

$$\exists x, y | P[Y = y | X = x] \neq P[Y = y]$$

- **Statistical association is merely an observational statement on the joint distribution of probability**
  - $P[Y = y | X = x]$  does not measure the *effect* of  $X$  on  $Y$

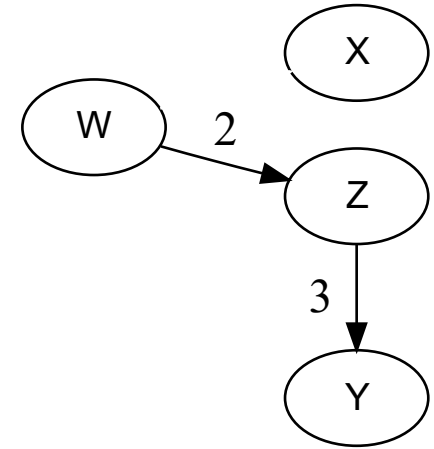


Weather ( $W$ ) influences ice cream sales ( $X$ ) and the number of swimmers ( $Z$ ), hence the number of drownings ( $Y$ ).

There is no directed path between  $X$  and  $Y$ , however  $X$  and  $Y$  are associated (red-dashed undirected edge), because of the unblocked path  $X \leftarrow W \rightarrow Z \rightarrow Y$ .

# What is Causation?

- **Causal association** flows along an unblocked *directed* path that starts in treatment  $X$  and ends in outcome  $Y$ , denoted the causal path
- Let  $do[X = x]$  represent the do-operator on  $X$ 
  - This is an intervention that sets the value of  $X$  to  $x$ , hence  $X$  is not influenced by any other variable
- **Definition:**  $X$  causes  $Y$  iff  $P[Y|do[X]] \neq P[Y]$
- Association implies causation only if all non-causal paths are blocked
- Causality is
  - an interventional (beyond observational) concept
  - asymmetric (directional)
  - sequential:  $X$  happens first, and then  $Y$  adapts

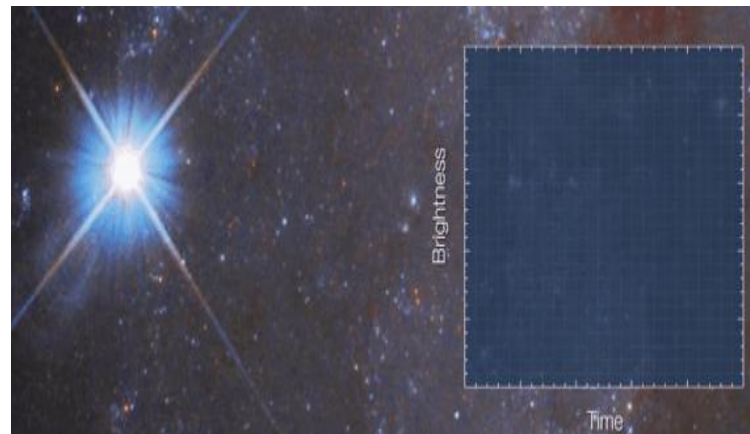


A do-operation on  $X$  removes arrow (1), because  $X$  is no longer a function of  $W$ , while keeping all other things equal (“ceteris paribus”). As a result, there is no unblocked path between  $X$  and  $Y$ , and  $P[Y|do[X]] = P[Y]$ .

# **A Cosmology Example: LCDM & Dark Energy**

# Supernovae Type Ia (SNe)

- SNe occur when a carbon-oxygen white dwarf approaches the Chandrasekhar limit (maximum mass of a stable white dwarf)
- SNe are useful because
  - their spectra are distinctive (unambiguous identification)
  - they are extremely bright (observable at far distances)
  - luminosity can be standardized (distance is measurable)
- These characteristics make SNe what cosmologists call *standardizable candles*
- **Implicit assumption: The brightness of SNe is primarily determined by their distance from us**



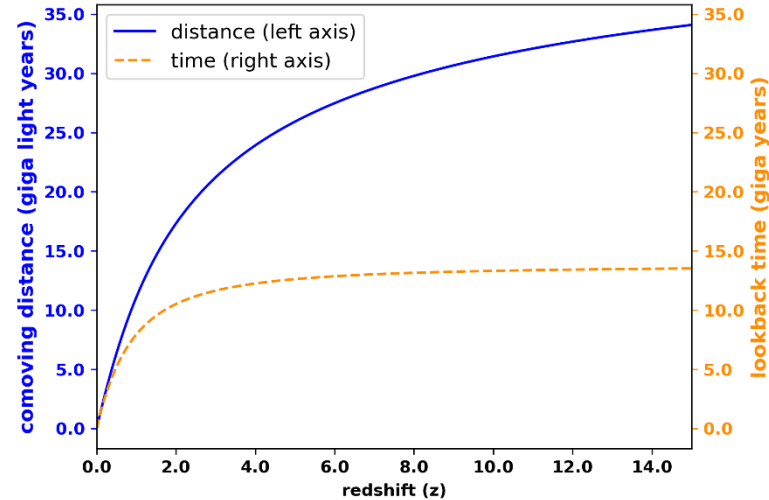
Light curve for a Supernova Type Ia, over the course of one year. SNe are easily identifiable, highly visible, and have standardized luminosity, making them ideal for measuring distances to far away host galaxies.

# Cosmological Redshift

- As space expands, every photon's wavelength is stretched along with it
- Redshift is a measure of how much the wavelength of light has been stretched while traveling through the expanding universe

$$z = \frac{\lambda_{obs} - \lambda_{emit}}{\lambda_{emit}}$$

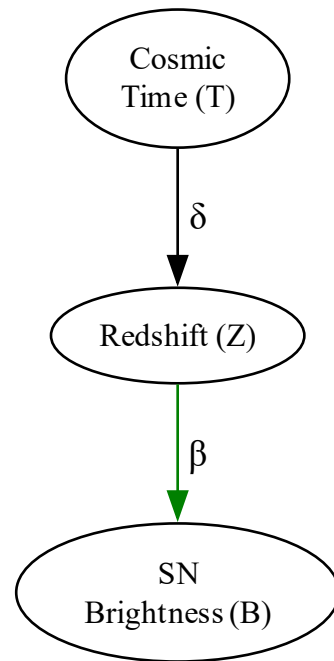
- If the object's distance is known, **redshift can be used to measure how much the universe has expanded** between emission and observation



For a given rate of expansion of the universe, redshift gives us the time light has been traveling through space. Given that light travels at a constant speed, we can use redshift to estimate distance (hence expected brightness).

# The Supernova Cosmology Project

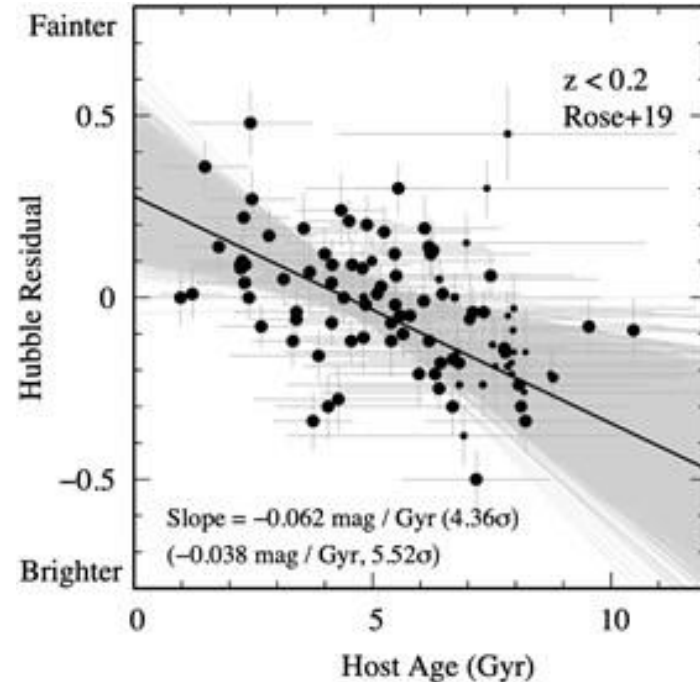
- The Supernova Cosmology Project (SCP) was established to measure the rate of expansion of the Universe, by comparing
  - the *observed* brightness of SNe with
  - the *expected* brightness of SNe given the observed redshift
- In 1998, the SCP found that SNe were dimmer than expected for a constant rate of expansion
  - SCP conjectured that “Dark Energy” may be responsible for the Universe’s accelerating expansion
- This discovery was named "Breakthrough of the Year for 1998" by Science Magazine
- In 2011, SCP researchers were awarded the Nobel Prize in Physics for their discovery



Causal graph implicitly assumed by the SCP, which led to their conjecture of “Dark Energy” and the LCDM model.

# The 2025 Shocker

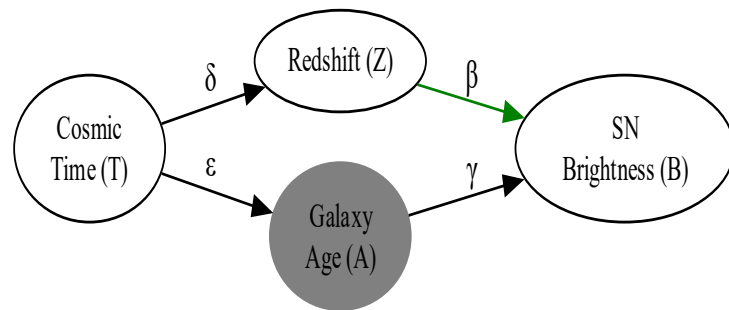
- In a paper published in November 2025 by the Royal Astronomical Society, [researchers](#) found that the age of the host galaxy is associated with
  - SNe Brightness, and
  - SNe Redshift
- This contradicts SCP's implicit assumption, that SNe are standardizable candles
  - Factors other than distance determine brightness of SNe
  - The discrepancy between observed and expected Brightness can be explained by something else than the accelerating expansion of the universe



Correlation between host galaxy age and Hubble Residual (a measure of luminosity) for SN host galaxies.

# Dark Energy As Confounder Bias

- Empirical evidence suggests that a more accurate causal graph is the one on the right
- Controlling for Galaxy\_Age blocks the backdoor path, and the 1998 observed-expected brightness discrepancy vanishes
  - the universe appears to be expanding at a constant speed
- **Confounder bias: IF confirmed, there is no need for “Dark Energy”, and the standard model of Cosmology (LCDM) is wrong**
- **For 27 years, cosmologist may have been searching for a mirage, a statistical illusion created by incorrect model specification**



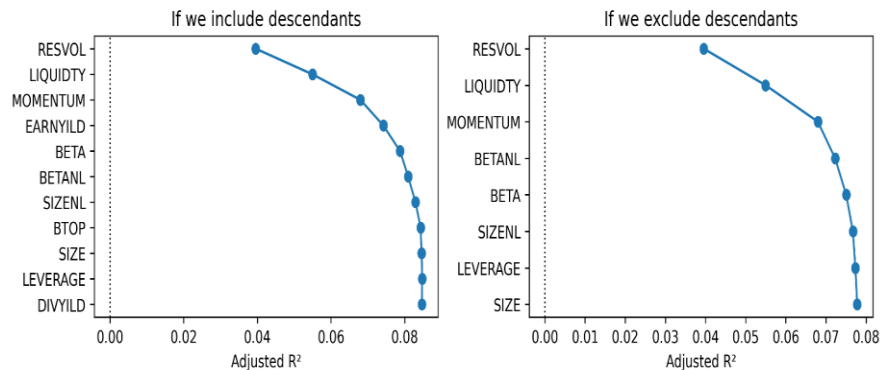
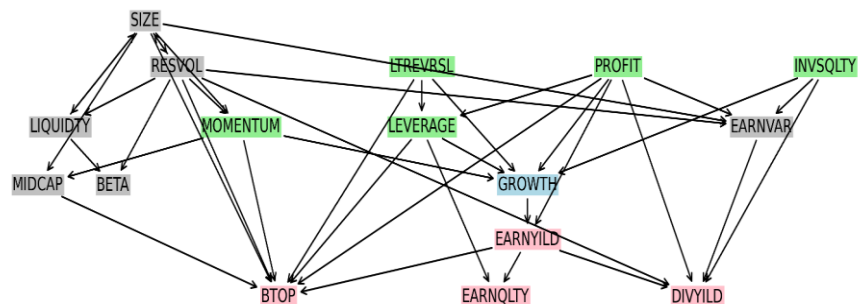
Not controlling for Galaxy\_Age biased the Brightness prediction, making it appear that observed SNe were farther than expected in a constant accelerating universe.

After controlling for Galaxy\_Age, the SNe data no longer imply acceleration and therefore do not require “Dark Energy.”

# **Investment Lessons from Cosmology**

# What Makes Cosmology & Finance Similar

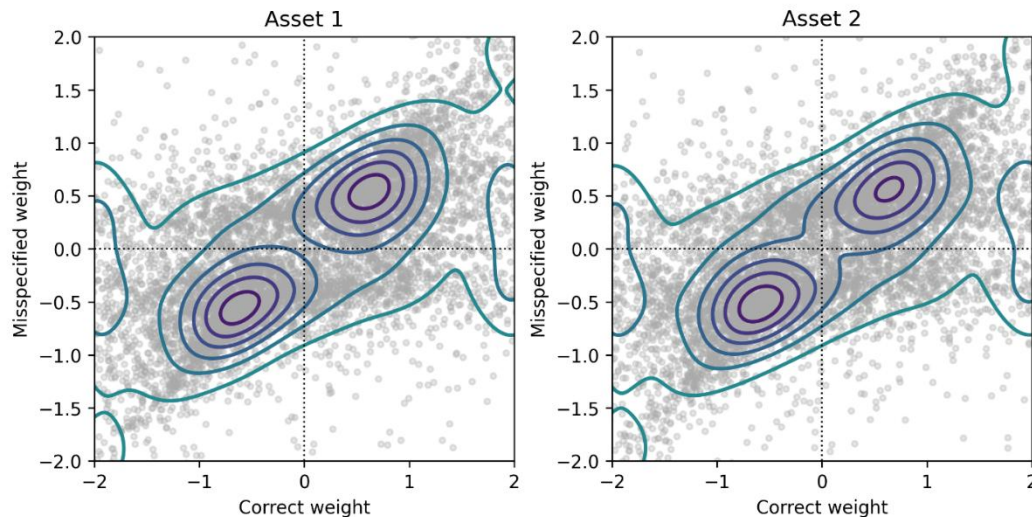
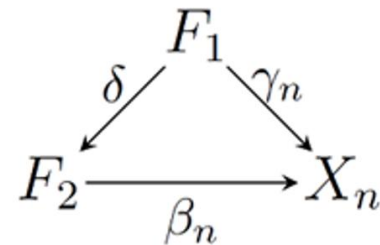
- Like Cosmologists, Financial researchers
  - rarely can conduct controlled experiments;
  - rely on historical simulations (regressions, backtests) to build models and strategies;
  - reach conclusions that rely on those models' implicit causal assumptions
- When those causal assumptions are wrong, the analysis leads to false strategies and likely investment losses
- While researchers cannot guarantee that causal assumptions are correct, they should always state them explicitly, so that they can be challenged separately from their simulations



The Factor Mirage: Investment strategies that purely rely on associations can achieve high performance in-sample, which cannot be monetized out-of-sample.

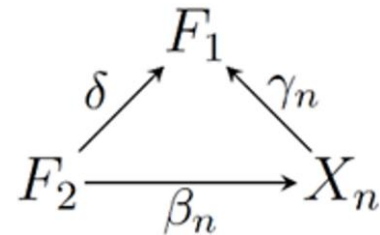
# Case 1: Failing to Control for a Confounder

- Applying a wrong backdoor adjustment set leads to “bad” strategies, where investors buy what they should have sold, and vice versa, even when investors have perfect knowledge of the mean and covariances.
- The figure below plots pairs  $(\omega_{true}^*, \omega_{miss}^*)$  when the model fails to control for a confounder
- For instance, when  $\beta_1 = \beta_2 = 1$ ,  $\delta = -0.5$ ,  $\gamma_1 = 1$ ,  $\gamma_2 = 1.5$ , then  $\omega_{true}^* = (3, -2)$ , whereas  $\omega_{miss}^* = (1.2, 0.8)$
- For additional details, [read this paper](#)

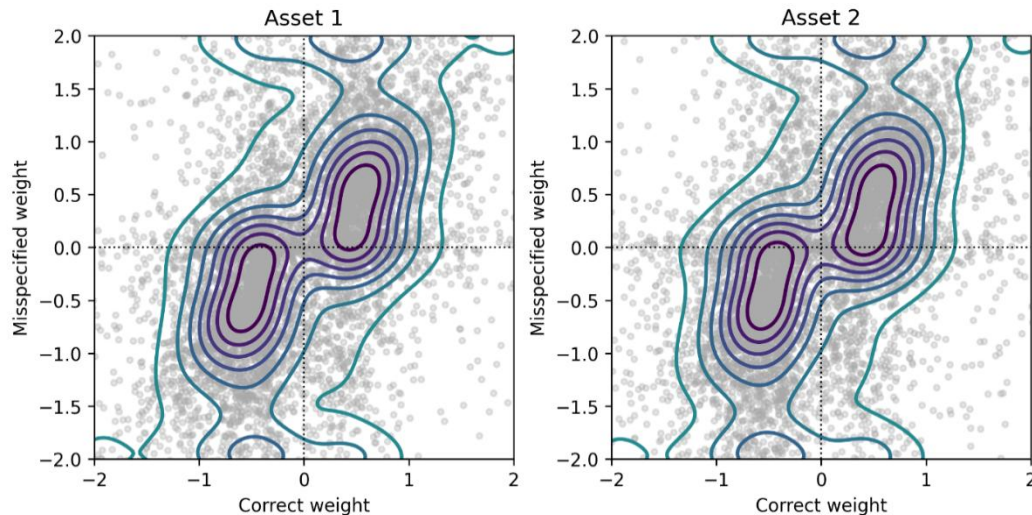


## Case 2: Controlling for a Collider

- Colliders are particularly dangerous, because they include scenarios where there is a reversal of *all* allocation signs
- The figure below plots pairs  $(\omega_{true}^*, \omega_{miss}^*)$  when the model control for a collider



- For instance, when  $\beta_1 = \beta_2 = 1$ ,  $\delta = 1.5$ ,  $\gamma_1 = 1.5$ ,  $\gamma_2 = -0.5$ , then  $\omega_{true}^* = (0.5, 0.5)$ , whereas  $\omega_{miss}^* = (-5, -3)$
- For additional details, [read this paper](#)



# Conclusions

# The Ranking of Empirical Evidence

- A scientific theory is a falsifiable statement of the form “ $X$  causes  $Y$  through mechanism  $M$ ”
- Scientific theories matter to investors because
  - causality is a necessary condition for investment efficiency
    - associational models misattribute risks and performance, thus preventing investors from building efficient portfolios
  - causal models enable counterfactual reasoning, hence the stress-testing of investment portfolios in a coherent and forward-looking manner
    - associational models cannot answer counterfactual questions, such as what would be the effect of  $Y$  on a not-yet-observed scenario  $X$ , thus exposing those relying on associations to black-swan events
- **The adoption of causal inference has the potential to transform investing into a truly scientific discipline**

| Type                                | Rigor     | Example                                                       |
|-------------------------------------|-----------|---------------------------------------------------------------|
| Randomized controlled trials        | Very high | Algo-wheel experiments                                        |
| Natural experiments                 | High      | Market-maker reaction to random spikes in order imbalance     |
| Simulated interventions             | Medium    | Estimate effect of HML using a causal graph                   |
| Econometric (observational) studies | Low       | Factor investing literature; backtested investment strategies |
| Case studies                        | Very low  | Broker report / analysis                                      |
| Expert opinion                      | Anecdotal | Investment guru's prediction                                  |

Hierarchy of evidence: **Pre-scientific** vs. **scientific evidence** in financial research.

# For More Information



Download for free

## **Causal Factor Investing**

(Cambridge University Press, 2023)

Available at:

<https://www.cambridge.org/core/elements/causal-factor-investing/9AFE270D7099B787B8FD4F4CBADE0C6E>

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